P Vs Np.hypotheses Evaluated

Hypothesis Evaluation Report

Stats

- Hypotheses evaluated: 44
- Support threshold: 0.62
- Refute threshold: 0.62
- Margin: 0.08
- Force resolution: False
- Coherence signals: 44
- Global coherence baseline: 0.313319

Verdict Counts

- inconclusive: 22
- supported: 22

Results

hypothesis_id	verdict	next_status	support	refute	confidence	action
phyp.p-vs-np.001	supported	supported	0.8093	0.0000	0.7398	promote-or-retest-with-ded
phyp.p-vs-np.033	supported	supported	0.8590	0.0000	0.8045	promote-or-retest-with-ded
phyp.p-vs-np.021	inconclusive	needs-testing	0.5462	0.0019	0.4342	run-another-abduce-deduce
phyp.p-vs-np.002	supported	supported	0.8028	0.0000	0.7316	promote-or-retest-with-ded
phyp.p-vs-np.022	inconclusive	needs-testing	0.5631	0.0027	0.4518	run-another-abduce-deduce
phyp.p-vs-np.023	inconclusive	needs-testing	0.5579	0.0029	0.4462	run-another-abduce-deduce
phyp.p-vs-np.003	supported	supported	0.8115	0.0000	0.7427	promote-or-retest-with-ded
phyp.p-vs-np.004	inconclusive	needs-testing	0.5692	0.0720	0.4404	run-another-abduce-deduce
phyp.p-vs-np.034	supported	supported	0.8379	0.0000	0.7767	promote-or-retest-with-ded
phyp.p-vs-np.035	supported	supported	0.8303	0.0000	0.7669	promote-or-retest-with-ded
phyp.p-vs-np.005	inconclusive	needs-testing	0.5934	0.0360	0.4752	run-another-abduce-deduce
phyp.p-vs-np.036	supported	supported	0.7090	0.0480	0.6008	promote-or-retest-with-ded
phyp.p-vs-np.007	supported	supported	0.6610	0.1078	0.5281	promote-or-retest-with-ded
phyp.p-vs-np.024	inconclusive	needs-testing	0.4589	0.0044	0.3462	run-another-abduce-deduce
phyp.p-vs-np.010	supported	supported	0.8206	0.0000	0.7543	promote-or-retest-with-ded

hypothesis_id	verdict	next_status	support	refute	confidence	action
phyp.p-vs-np.011	supported	supported	0.6693	0.0360	0.5589	promote-or-retest-with-ded
phyp.p-vs-np.025	inconclusive	needs-testing	0.4662	0.0048	0.3532	run-another-abduce-deduce
phyp.p-vs-np.012	supported	supported	0.7621	0.0000	0.6805	promote-or-retest-with-ded
phyp.p-vs-np.013	inconclusive	needs-testing	0.6118	0.0000	0.5049	run-another-abduce-deduce
phyp.p-vs-np.016	supported	supported	0.7351	0.0000	0.6475	promote-or-retest-with-ded
phyp.p-vs-np.017	inconclusive	needs-testing	0.5439	0.1272	0.4011	run-another-abduce-deduce
phyp.p-vs-np.018	inconclusive	needs-testing	0.5760	0.0360	0.4568	run-another-abduce-deduce
phyp.p-vs-np.037	supported	supported	0.7262	0.0264	0.6281	promote-or-retest-with-ded
phyp.p-vs-np.038	supported	supported	0.7237	0.0312	0.6236	promote-or-retest-with-ded
phyp.p-vs-np.039	supported	supported	0.8448	0.1858	0.7151	promote-or-retest-with-ded
phyp.p-vs-np.040	supported	supported	0.7122	0.2019	0.5553	promote-or-retest-with-ded
phyp.p-vs-np.041	supported	supported	0.7274	0.0000	0.6382	promote-or-retest-with-ded
phyp.p-vs-np.042	supported	supported	0.8848	0.0000	0.8389	promote-or-retest-with-ded
phyp.p-vs-np.043	supported	supported	0.7045	0.0264	0.6025	promote-or-retest-with-ded
phyp.p-vs-np.044	supported	supported	0.7495	0.0000	0.6650	promote-or-retest-with-ded
phyp.p-vs-np.026	inconclusive	needs-testing	0.4723	0.0061	0.3588	run-another-abduce-deduce
phyp.p-vs-np.027	inconclusive	needs-testing	0.4238	0.0070	0.3126	run-another-abduce-deduce
phyp.p-vs-np.028	inconclusive	needs-testing	0.4512	0.0071	0.3383	run-another-abduce-deduce
phyp.p-vs-np.029	inconclusive	needs-testing	0.4203	0.0074	0.3093	run-another-abduce-deduce
phyp.p-vs-np.030	inconclusive	needs-testing	0.4556	0.0079	0.3423	run-another-abduce-deduce
phyp.p-vs-np.031	inconclusive	needs-testing	0.4362	0.0083	0.3239	run-another-abduce-deduce
phyp.p-vs-np.006	inconclusive	needs-testing	0.5855	0.0180	0.4716	run-another-abduce-deduce
phyp.p-vs-np.008	inconclusive	needs-testing	0.5749	0.1402	0.4286	run-another-abduce-deduce
phyp.p-vs-np.009	supported	supported	0.7157	0.0000	0.6242	promote-or-retest-with-ded
phyp.p-vs-np.014	inconclusive	needs-testing	0.5282	0.1253	0.3863	run-another-abduce-deduce
phyp.p-vs-np.015	supported	supported	0.6884	0.0360	0.5807	promote-or-retest-with-ded
phyp.p-vs-np.032	inconclusive	needs-testing	0.4201	0.0092	0.3088	run-another-abduce-deduce
phyp.p-vs-np.019	inconclusive	needs-testing	0.5287	0.0360	0.4080	run-another-abduce-deduce
phyp.p-vs-np.020	inconclusive	needs-testing	0.6058	0.1413	0.4599	run-another-abduce-deduce

Mathematical Readout

phyp.p-vs-np.001 [bridge-link]

- Verdict: supported (support=0.8093, refute=0.0000, confidence=0.7398)
- Antecedent schema: nonuniform-circuit-exponential-lower-bound
- Target consequence: NP-not-in-P/poly
- Preserved invariant: MCSP complexity profile in nonuniform-circuits
- Barrier scope: none
- Alignment signals: antecedent=0.3069, consequence=0.5070, invariant=0.3050, barrier=0.0000
- Key reasons: structural signature overlap between support and target; symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector anal-

ysis score for support->target pair

phyp.p-vs-np.033 [definition-introduction]

- Verdict: supported (support=0.8590, refute=0.0000, confidence=0.8045)
- Antecedent schema: nonuniform-circuit-exponential-lower-bound
- Target consequence: NP-not-in-P/poly
- Preserved invariant: MCSP complexity profile in nonuniform-circuits
- Barrier scope: none
- Alignment signals: antecedent=0.3069, consequence=0.5070, invariant=0.3050, barrier=0.0000
- Key reasons: conceptual_kind=definition-introduction; connector support for conceptual bridge present; nontrivial structured semantic overlap between support and target contexts; conceptual move preserves explicit invariant: MCSP complexity profile in nonuniform-circuits

phyp.p-vs-np.021 [intervention-focus]

- Verdict: inconclusive (support=0.5462, refute=0.0019, confidence=0.4342)
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.0000, barrier=0.0000
- Key reasons: high priority inherited from connector hypotheses; strong cross-component potential

phyp.p-vs-np.002 [bridge-link]

- Verdict: supported (support=0.8028, refute=0.0000, confidence=0.7316)
- Antecedent schema: nonuniform-circuit-exponential-lower-bound
- Target consequence: NP-not-in-P/poly
- Preserved invariant: MCSP complexity profile in nonuniform-circuits
- Barrier scope: none
- Alignment signals: antecedent=0.3069, consequence=0.5068, invariant=0.3050, barrier=0.0000
- Key reasons: structural signature overlap between support and target; symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector analysis score for support->target pair

phyp.p-vs-np.022 [intervention-focus]

- Verdict: inconclusive (support=0.5631, refute=0.0027, confidence=0.4518)
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.0000, barrier=0.0000
- Key reasons: high priority inherited from connector hypotheses; strong cross-component potential

phyp.p-vs-np.023 [intervention-focus]

- Verdict: inconclusive (support=0.5579, refute=0.0029, confidence=0.4462)
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.0000, barrier=0.0000
- Key reasons: high priority inherited from connector hypotheses; strong cross-component potential

phyp.p-vs-np.003 [bridge-link]

- Verdict: supported (support=0.8115, refute=0.0000, confidence=0.7427)
- Antecedent schema: proof-system-characterization
- Target consequence: TAUT-proof-system-characterization
- Preserved invariant: TAUT complexity profile in Cook-Reckhow-proof-system
- Barrier scope: none
- Alignment signals: antecedent=0.5104, consequence=0.5795, invariant=0.6671, barrier=0.0000
- Key reasons: structural signature overlap between support and target; symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector analysis score for support->target pair

phyp.p-vs-np.004 [bridge-link]

- Verdict: inconclusive (support=0.5692, refute=0.0720, confidence=0.4404)
- Antecedent schema: lower-bound
- Target consequence: nonuniform-circuit-lower-bound
- Preserved invariant: consequence schema nonuniform-circuit-lower-bound
- Barrier scope: none
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.2567, barrier=0.0000
- Key reasons: structural signature overlap between support and target; symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector analysis score for support->target pair

phyp.p-vs-np.034 [definition-introduction]

- Verdict: supported (support=0.8379, refute=0.0000, confidence=0.7767)
- Antecedent schema: nonuniform-circuit-exponential-lower-bound
- Target consequence: NP-not-in-P/poly
- Preserved invariant: MCSP complexity profile in nonuniform-circuits
- Barrier scope: none
- Alignment signals: antecedent=0.3069, consequence=0.5068, invariant=0.3050, barrier=0.0000

- Key reasons: conceptual_kind=definition-introduction; connector support for conceptual bridge present; nontrivial structured semantic overlap between support and target contexts; conceptual move preserves explicit invariant: MCSP complexity profile in nonuniform-circuits

phyp.p-vs-np.035 [definition-introduction]

- Verdict: supported (support=0.8303, refute=0.0000, confidence=0.7669)
- Antecedent schema: proof-system-characterization
- Target consequence: TAUT-proof-system-characterization
- Preserved invariant: TAUT complexity profile in Cook-Reckhow-proof-system
- Barrier scope: none
- Alignment signals: antecedent=0.5104, consequence=0.5795, invariant=0.6671, barrier=0.0000
- Key reasons: conceptual_kind=definition-introduction; connector support for conceptual bridge present; nontrivial structured semantic overlap between support and target contexts; conceptual move preserves explicit invariant: TAUT complexity profile in Cook-Reckhow-proof-system

phyp.p-vs-np.005 [bridge-link]

- Verdict: inconclusive (support=0.5934, refute=0.0360, confidence=0.4752)
- Antecedent schema: lower-bound
- Target consequence: nonuniform-circuit-superpolynomial-lower-bound
- Preserved invariant: nonuniform-circuit-lower-bound witness complexity
- Barrier scope: none
- Alignment signals: antecedent=0.0000, consequence=0.6139, invariant=0.3827, barrier=0.0000
- Key reasons: structural signature overlap between support and target; symbol-role alignment indicates semantic compatibility; theorem-family embedding similarity indicates structural proximity; high connector analysis score for support->target pair

phyp.p-vs-np.036 [definition-introduction]

- Verdict: supported (support=0.7090, refute=0.0480, confidence=0.6008)
- Antecedent schema: lower-bound
- Target consequence: nonuniform-circuit-lower-bound
- Preserved invariant: consequence schema nonuniform-circuit-lower-bound
- Barrier scope: none
- Alignment signals: antecedent=0.0000, consequence=0.0000, invariant=0.2567, barrier=0.0000
- Key reasons: conceptual_kind=definition-introduction; connector support for conceptual bridge present; nontrivial structured semantic overlap between support and target contexts; conceptual move preserves explicit invariant: consequence schema nonuniform-circuit-lower-bound